

TABLE I. Finite-window design recipe implied by Theorem 1. Each step cites the statement that licenses it; the recipe prescribes modality only inside the declared window I .

Given finite $d \geq 2$ and $m \geq 1$, a window $I = [\tau, T] \subset (0, \infty)$, and target times $\tau < t_1 < \dots < t_m < T$:

(1) Transport. Choose $(\gamma, z_0, \bar{z})$ with $z_0 \neq \bar{z}$ so that the midpoint mean $\mu(t) = \bar{z} + (z_0 - \bar{z})e^{-\gamma t}$ traverses the design range monotonically with $x'(t)$ bounded away from zero on I [Eq. (9)]. On a compact window $x' > 0$ holds automatically; the quantitative point is to keep $T \lesssim$ a few γ^{-1} so $x'(t_m)$ is not exponentially small (late peaks widen as $\sigma/x'(t_j)$).

(2) Support. Place the slab centres at $c_j = x(t_j)$, i.e. physical centres $\mu(t_j)$ [Eq. (9)].

(3) Width. Choose ε (and ρ) so that σ of Eq. (14) obeys $\min_j(c_{j+1} - c_j) \geq C\sigma$ with $C \approx 2.3$. At this separation the crossover analysis [Eqs. (16)–(17)] yields one valley per adjacent gap with nonadjacent components exponentially small; strongly unbalanced weights shift the crossover by $\sigma^2 \log(w_j/w_{j+1})/(c_{j+1} - c_j)$ and eat into the margin. Theorem 1 guarantees only that some $\varepsilon < \varepsilon_0$ works; $C \approx 2.3$ is a practical proxy, slightly above the equal-weight two-Gaussian bimodality threshold 2σ .

(4) Verify the free clock. Evaluate $G_{\varepsilon, w}(t) = A_\varepsilon(t)H_{\sigma, w}(x(t))$ [Eqs. (13)–(14)] on I —a cheap one-dimensional computation; the contact factor $c_{d, \varepsilon}(t)$ follows from Gaussian and wrapped-Gaussian marginals—and confirm exactly m nondegenerate maxima and $m - 1$ nondegenerate minima, with clear root separation, curvature signs, and endpoint slopes.

(5) Budget. Choose $B < B_0(\varepsilon)$ so that $f_{B, \varepsilon, w}/B$ inherits the complete stationary signature of $G_{\varepsilon, w}$ [Theorem 1; Eqs. (6), (19)]. The theorem asserts that $B_0(\varepsilon) > 0$ exists but provides no useful lower bound: take B from an empirically measured depletion-threshold curve $B_0(\varepsilon)$ —a separate numerical validation gate outside this theorem (Sec. IV)—and re-verify f directly at the chosen B .

Caveat (quantifier order). The choices are sequential: the geometry and admissible scales may depend on (d, m) ; one first fixes $\varepsilon < \varepsilon_0$ and only then $B < B_0(\varepsilon)$. Nothing is claimed about modality outside I , about an event-mass floor, or about uniformity of ε_0 , B_0 in d or m (Secs. IV and V).

PREVIEW: DESIGN RECIPE BOX SNIPPET

Table I summarizes the construction as a checkable procedure.